A portable local framework for gene-structure and expression queries in underrepresented plant species

Article type: Original Article (submission target: *Annals of the New York Academy of Sciences*)

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Abstract

Plant gene annotations, genome sequences, and expression matrices are distributed across repositories, often without a unified species-specific query interface. We present PhytoScope, a local desktop framework that imports user-supplied genome annotations, matching sequences, and expression matrices into a locus-centered workspace. We evaluated portability, data fidelity, and reproducibility using reference annotations from four established plant systems and three additional plants for which the evaluated workflow relies on general public archives rather than a dedicated query portal. Across eight source-derived annotation inventories, 4,641,758 recognized gene, transcript, exon, coding-sequence, and untranslated-region tuples matched exactly, with no missing or extra tuples. In three additional plant assemblies, full-annotation import and locus search were tested; source-derived sequence gold standards covered five coding loci per assembly. A public rice-bean RNA-seq dataset was re-quantified against the exact RefSeq transcript release: all 34,914 resulting gene identifiers matched that release's annotation. Because reads came from genotype IC426787 and the reference assembly represents cultivar VRB3, this is an identifier-compatibility and data-import result, not genotype-matched expression biology. The framework integrates sequence extraction, expression display, and project persistence, but current evidence does not establish usability, time savings, biological discovery, or universal format compatibility.

Keywords: plant genomics; non-model species; genome annotation; gene expression; data portability; reproducible research; bioinformatics software

Graphical abstract summary (≤70 words): PhytoScope imports user-supplied plant annotation, genome sequence, and expression data into a local, recoverable locus workspace. Source-derived checks across seven plant species test annotation portability and sequence reconstruction. Rice-

bean RNA-seq reads quantified against an exact RefSeq transcript release validate expression-matrix import and identifier reconciliation, while genotype differences constrain biological interpretation.

Introduction

Plant genome resources have expanded rapidly, but the availability of sequence data is not equivalent to the availability of a practical, species-specific gene information system. A researcher may have to combine an annotation file, a reference FASTA, expression tables, and sample metadata from separate sources before answering routine locus questions. Mature resources such as TAIR, Gramene, MaizeGDB, SoyBase, and Phytozome provide valuable infrastructure for selected plant models and crops [1–5]. Genome browsers such as JBrowse and IGV provide general visualization capabilities [6–8]. These resources solve important problems, but coverage, integration, and available query functions vary by species and project. For plants without a dedicated query portal, researchers still need a reliable way to inspect public files without first building and maintaining a species-specific web database.

This gap is particularly relevant to underutilized crops, medicinal plants, ornamentals, and locally studied taxa. Public archives increasingly contain genome assemblies and gene models for such plants, while analysis often remains fragmented across command-line tools and spreadsheets. General file standards such as GFF3 enable reuse, yet real annotations differ in sequence naming, parent-child structure, optional attributes, and treatment of stop codons and untranslated regions. Downstream sequence extraction additionally depends on using the matching assembly and a clear coordinate convention [9,10]. A portable tool therefore needs to expose data provenance and missing information rather than silently fill gaps or imply that an annotation is a validated biological truth.

PhytoScope is a local desktop framework designed for this setting. Its premise is that the same query and visualization software can be reused across species when users provide compatible annotation, sequence, and expression files. This approach differs from a curated species database: PhytoScope does not supply a canonical genome release, functional curation, community annotation, or species-specific literature. It instead connects user-provided files through a locus-centered interface, supports gene and transcript inspection and sequence export, and can preserve selected inputs and analyses in a portable project. This complements, rather than replaces, database and browser infrastructure.

We asked whether one unmodified application could (1) import and query public annotations across diverse plant data sets, including three additional plants outside the established four-species regression cohort; (2) reproduce source-derived structural and sequence records; and (3) retain links among loci, user-supplied expression values, analyses, and project state. We evaluated these properties with independent coordinate and sequence checks, a narrow scripted comparator, module-level test oracles, and a public RNA-seq re-quantification. The evaluation is intentionally technical. It does not test researcher usability, time savings, scientific discovery, or a biological hypothesis. In particular, public expression reads from rice bean were generated from IC426787, whereas the tested RefSeq assembly is VRB3. Our results thus assess importing and reconciling expression data to an exact annotation release, not genotype-matched expression biology. This distinction also defines the intended contribution: a reproducible query layer that works with local data files, not a new reference annotation, a centrally curated database, or an analytical method that changes the biological interpretation of its inputs. We therefore evaluated separate technical claims independently: whether records can be imported and searched, whether structures and selected sequences reproduce their source definitions, and whether expression IDs

can be joined to the matching release. This decomposition makes it possible to identify precisely where source completeness, reference matching, or user metadata limits a downstream result.

Materials and Methods

Software and data model

PhytoScope is a Java desktop application with a JavaFX interface. The evaluated workflow accepts plant annotation files in GFF3 or GTF and, where sequence extraction is required, a corresponding genomic FASTA. It parses recognized gene, transcript, exon, CDS, and UTR features and their parent relationships into a searchable local model. The interface supports sequence accession discovery, gene identifier and coordinate search, gene/transcript structure display, sequence export, expression-matrix loading, expression visualization, and locus-linked research operations. Research operations include promoter or custom-interval extraction, sequence quality checks, protein similarity comparison, domain-to-CDS/genomic mapping, and explicit sample grouping. The application records file references, selected locus and view state, inputs, and analysis outputs in a recoverable project. It does not infer gene function, orthology, regulatory causality, or differential expression. Feature types outside the recognized model remain outside the validation claim.

Plant data sets and portability cohort

The reference cohort comprised *Arabidopsis thaliana* TAIR10/Araport11, *Oryza sativa* IRGSP-1.0, *Zea mays* B73 NAM5, and *Glycine max* v2.1. It was used for full annotation and cross-format regression testing. The portability cohort comprised *Vigna umbellata* RefSeq assembly GCF_029865625.1 (Vumbellata_1.0; cultivar VRB3; annotation release GCF_029865625.1-RS_2026_09), *Andrographis paniculata* GCF_009805555.1 (Annotation Release 100), and *Hibiscus syriacus* GCF_006381635.1 (Annotation Release 100). Full GFF3

files were retrieved from NCBI and checked against release checksums. Rice bean was additionally tested with the corresponding GTF and chromosome FASTA. For *Andrographis* and *Hibiscus*, tests included public scaffold sequences for selected loci. Assembly, annotation, and expression availability are release-specific; we do not assume that every species provides all data types or a dedicated portal.

Annotation fidelity and sequence gold standards

For four reference species, source inventories were generated independently of PhytoScope from pinned annotation records. Eight inventories covered bundled Arabidopsis annotation, full Ensembl Plants release-62 annotation for Arabidopsis, and source excerpts and full annotations for rice, maize, and soybean. Comparisons used explicit parent relationships, strand, and exact 1-based inclusive coordinates for recognized gene, transcript, exon, CDS, and UTR records. Missing and extra tuples were counted separately. A second comparison used `gffutils` version 0.13 on release-62 excerpts containing 10 Arabidopsis, 24 rice, 20 maize, and 40 soybean loci. Three fresh-process trials per species alternated tool order; both outputs were checked against the source-derived tuples. This comparison measures annotation hierarchy extraction only. Import time and SQLite index-build time are different operations and were not interpreted as a speed ranking.

For the three portability-cohort species, a separate builder reconstructed sequences directly from source GFF3 intervals and matching accession FASTA, without calling the PhytoScope parser. Five coding loci were selected per species, including coding transcript models. Frozen records capture gene bounds, strand, exon and CDS lengths, and SHA-256 digests for genomic span, spliced transcript, and CDS. Rice-bean locus `LOC124833352` received additional checks against RefSeq CDS/protein records and annotation formats. Its GFF3 CDS includes a three-base

terminal stop codon that the corresponding GTF represents separately as `stop_codon`; these representations were kept distinct and translated protein sequence was compared. UTR absence in this source was not interpreted as evidence that a transcript has no UTR; any computed interval was labeled inferred.

Expression data and matrix reconciliation

We used three paired-end 5-days-post-anthesis libraries from the public rice-bean BioProject PRJNA765494 [16]: SRR16122599, SRR16122608, and SRR16122607. The source genotype was IC426787. Reads were re-quantified with Salmon 2.8.0 using selective alignment against the RefSeq RNA FASTA associated with assembly GCF_029865625.1 (54,178 transcript sequences; pinned SHA-256 in the manifest). Transcript-to-gene mapping used parent relationships in the matching RefSeq GFF3. Salmon `--geneMap` generated a gene TPM matrix; no tximport length correction or differential-expression analysis was performed. FASTQ sizes and MD5 hashes, input checksums, quantification settings, per-run reports, matrix hash, and sample labels are recorded in the repository manifest. Matrix validation checked parsed cells and exact gene-ID overlap with imported GFF3 gene/pseudogene IDs. The genotype mismatch was retained as a stated interpretive limitation.

Independent research-module checks

Promoter extraction was tested for eight TAIR10 loci on both strands, each using a 2,200-bp interval, against direct reference-FASTA slicing and reverse complementation. bedtools 2.31.1 `getfasta -s` [17] served as an independent sequence comparator. Four Arabidopsis RBCS CDS models were checked against independent length, frame, base ambiguity, start, and stop calculations. Within-family protein-alignment metrics were compared to a test-side Smith–Waterman implementation; external BLASTP 2.12.0+ [18] comparisons used shared thresholds

on RBCS positives and unrelated-family negative controls. Expression grouping arithmetic was checked against Expression Atlas E-MTAB-7978 values [19], including measured zeros. Sixteen domain/deletion mappings across 13 genes were reconstructed with an independent Python validator. These are bounded module tests, not general sensitivity or specificity estimates. JavaFX regression tests exercised visible actions and outputs in the five research tabs and project save, replacement, and restoration. Three-platform CI ran on Ubuntu, Windows, and macOS; reports are linked in the repository evidence package.

Reproducibility and claim criteria

Source URLs, accessions, release names, checksums, parameters, commands, frozen gold sets, and CI reports are retained in the repository. Tests were designed so that outputs could be compared to source-derived records without relying on the application parser. We report exact agreement only for the tested feature classes, files, releases, and selected loci. We did not conduct participant testing or a matched end-to-end comparison against manual and browser workflows. Consequently, no claim about usability, human efficiency, user error, discovery rate, or superiority is made.

Results

One local workflow queried multiple plant annotation releases

The same parser, search model, and desktop query workflow were used for all seven taxa; no species-specific parser branch or species-specific query code was required for the evaluated cases. Four established reference systems exercised chromosome and scaffold naming, GFF3/GTF, and full-annotation loading. In the three additional plant assemblies, the application imported complete GFF3 annotations and supported searches and structure views for selected loci, including targets on unplaced scaffolds. The portability cases demonstrate a file-driven

workflow across diverse plant inputs, not universal compatibility with arbitrary annotations. User-supplied matching sequence remains necessary for sequence extraction, and the application cannot recover information absent from annotation or metadata.

The rice-bean case illustrates why exact source representation matters. The assembly contains 11 assembled chromosomes and 221 unplaced scaffolds. Its tested GFF3 covered 232 sequence IDs, whereas the matching GTF covered 99. The tested locus `LOC124833352` is on chromosome accession NC_142174.1 and has a three-exon coding model. For this locus, PhytoScope reproduced the source coordinates and sequence products. The GFF3 CDS spans 1,983 nt with the stop codon included; in the GTF, the CDS spans 1,980 nt and the stop codon is a separate three-base feature. The proteins translate identically. The annotation has no explicit UTR rows at this locus, so any UTR interval displayed from transcript/CDS boundaries is marked as inferred. This behavior preserves an important distinction between source-recorded and derived information.

Source-derived structural tuples matched exactly in the reference cohort

Across eight annotation inventories, the source-to-model audit compared 4,641,758 recognized structural tuples. No tuple was missing or extra. The full-annotation total included 791,557 Arabidopsis, 550,921 rice, 1,161,691 maize, and 1,324,199 soybean tuples; source excerpts and bundled Arabidopsis data comprised the remaining inventories. Exact agreement applied to recognized feature classes and the explicit parent, strand, and coordinate fields tested. It does not validate unsupported annotation feature types or biological correctness of source gene models. The scripted `gffutils` comparison independently agreed with PhytoScope and the source-derived inventory for all four release-62 excerpts in three alternating-order runs per species. This validates a narrow annotation hierarchy and coordinate extraction task. Because the tools

performed different setup operations and the benchmark was not a matched full-workflow or controlled hardware study, the recorded timings do not support a performance ranking. The additional three-platform CI tests establish that the documented software tests passed in the recorded Ubuntu, Windows, and macOS environments; they do not establish compatibility with every operating system configuration or arbitrary user file. The software test suite includes automated checks for coordinate navigation, synchronized locus state, visible research-tab outputs, and project restoration. These checks address whether the implemented interface actions produce the expected recorded outputs. They cannot answer whether a first-time researcher understands the controls, can select appropriate input files without help, or completes a biological task more quickly than with another workflow. Those questions require a study with representative participants and prespecified task outcomes.

Selected sequence products matched independent reconstruction in three additional plants

The frozen non-model sequence gold set contains 25 coding-transcript records: five selected loci in each of the three portability-cohort species. For each record, source GFF3 and matching accession FASTA were used to independently calculate gene bounds, strand, exon/CDS lengths, and sequence hashes. PhytoScope exports and locus metadata matched the frozen values. Rice bean was tested more deeply at one locus against a RefSeq CDS/protein record, including the distinct GFF3 and GTF stop-codon conventions. These checks add sequence-level evidence beyond import and search, but five loci per species cannot represent genome-wide sequence accuracy or every annotation edge case. The tested loci span assembled chromosomes and unplaced sequence accessions in the cohort; this is useful because identifiers on unplaced scaffolds are common in less curated resources and are not safely handled by assuming a short list of chromosome names. However, selected success cases cannot establish performance across

all scaffolds or all structurally unusual models. The number of tested loci is therefore reported explicitly, and the full annotation import result is kept separate from sequence-gold coverage.

Public rice-bean reads validated matrix import and exact identifier reconciliation

Salmon quantified the three IC426787 libraries against the pinned GCF_029865625.1 RefSeq transcript set. The gene-level output contained 34,914 rows. Every output identifier matched a gene or pseudogene identifier in the matching GFF3 (34,914/34,914; zero unmatched); the matrix contained no missing or invalid cells, and the three TPM columns summed to approximately 1,000,000 within rounding. Salmon reported IU library format and a compatible-fragment ratio of 1.0 for each run. These values establish successful re-quantification, parsing, and exact-release identifier reconciliation. The matrix contains targets retained by the Salmon gene map, not every annotated model; an absent row cannot be interpreted as zero expression. IC426787 reads were quantified against the VRB3 reference, so these results are not genotype-matched expression biology, a differential-expression analysis, or evidence for a rice-bean expression pattern.

Research functions produced independently checked outputs within bounded panels

Eight strand-aware 2,200-bp promoter intervals matched direct FASTA slicing and reverse-complement calculations; bedtools independently returned the same eight sequences. Four RBCS CDS length/QC records matched independent metrics. In the Arabidopsis RBCS panel, BLASTP and PhytoScope agreed on 12 of 12 non-self family pairs and eight of eight unrelated-family negative-control pairs. The broader four-species mature-RBCS panel likewise showed 20/20 pairwise classification agreement. Since the panel covers one protein family and is small, it is a smoke test for similarity outputs, not an orthology benchmark. Sixteen domain/deletion coordinate mappings in 13 genes matched an independent Python reconstruction. The expression

grouping check reproduced the E-MTAB-7978 arithmetic for the selected groups and retained measured zero values; it did not test differential expression. JavaFX tests verified visible outputs and project save, replacement, and restoration across research functions. Together these checks support technical operation and selected calculations, while leaving broad module accuracy, real-world task performance, and biological conclusions outside the evidence.

Discussion

The evaluation supports a bounded proposition: when researchers have public annotation and matching sequence files, a single local framework can provide consistent locus search, structural inspection, sequence extraction, expression display, and selected locus-linked analyses across several plant data sets without species-specific query code for the tested cases. The framework is useful as a software architecture precisely because its unit of adaptation is the data set and its manifest rather than a branch of application code for every species. A project can be initialized from compatible files, and a saved workspace can then preserve the association between a selected locus, view state, and analysis outputs. This property is observable from the implementation and automated project round-trip checks. It does not establish that researchers prefer this organization or that it reduces the number of steps in their work. This matters for taxa whose data are available as general archive files but for which a project may not have a dedicated, maintained query portal. It is a framework for bringing data together locally, not a replacement for curated resources such as TAIR, Gramene, MaizeGDB, SoyBase, Phytozome, or general genome browsers [1–8]. Those resources provide curated content and capabilities beyond the scope of PhytoScope.

The evidence combines independent checks at different scales. Full annotation audits test large numbers of recognized structural tuples in four reference species. Public plant cases test file

portability and selected sequence products in three additional assemblies. Rice bean adds a complete public-read quantification and exact ID join to a pinned annotation release. Module tests check particular calculations against direct reconstruction or external tools. These layers answer different questions and should not be collapsed into a single “accuracy” score. Structural tuple agreement does not establish the truth of the source models; selected sequence checks do not prove genome-wide fidelity; and a small homology panel does not establish orthology. The distinction follows good computational practice: define the task, reference, input version, and failure boundary for each validation [11–13].

The public-expression exercise demonstrates both practical utility and an important limitation. The source files are public and reproducible; the generated matrix can be loaded and its 34,914 gene identifiers reconciled to the exact release used by the software. Yet the sequenced genotype, IC426787, is different from the VRB3 cultivar represented by the tested reference assembly. Re-quantification resolves the mismatch between the original expression models and the current tested annotation at the identifier level, but it cannot remove sequence polymorphism or make this a genotype-matched expression study. No differential-expression test was conducted, and the output matrix is not length-corrected by tximport. Expression data also require user-supplied units, sample labels, experimental design, and replicate metadata; annotation coordinates alone cannot provide these. Such explicit boundaries are essential for interpreting public RNA-seq resources [14,15]. The comparison also illustrates a general data-integration constraint: the read archive records the biological sample and its deposited labels, while a reference assembly records a particular genome source and version. Exact identifier matching is a useful computational check, but it does not demonstrate that the samples and assembly share a genotype, nor that all polymorphic loci are represented equivalently. Users must review the

accession-level provenance and not use a successful join as a proxy for experimental design validation.

PhytoScope's local project model can preserve selected inputs, analysis outputs, and locus state in a recoverable object. This provides a technical route to reproduce a recorded project, but project persistence should not be conflated with FAIR compliance or independent reproducibility of every external source. Reproduction still depends on the user retaining or reacquiring the exact source files, knowing their provenance, and supplying compatible sequence and metadata. The repository therefore records version identifiers, checksums, parameters, test gold standards, and run reports, in line with established reproducible computational workflow recommendations [11–13,20].

Several limitations define the next research steps. First, portability testing of three additional plants is broad but shallow: full import and search are complemented by only five selected sequence-gold loci per species. More annotation dialects, optional-field omissions, unusual parent relationships, organellar genomes, and larger assemblies should be tested. Second, the exact tuple audit concerns selected feature types. Unsupported or ambiguous records should remain visible to users, and a formal compatibility matrix should distinguish parsed, ignored, and inferred data. Third, module-level validation is uneven. Promoter extraction has a small independent external comparison; protein similarity is one-family smoke testing; domain mapping has a bounded set of existing models; expression grouping uses single-assay values. Curated multi-family homology and domain sets and a multi-replicate matrix gold standard are needed before broader accuracy statements. Fourth, current comparison with `gffutils` covers scripted annotation extraction only. A browser is designed to excel at navigation and visualization, and a general-purpose system may offer track configuration, remote hosting, and

data types outside this application's scope. A fair comparison should therefore not treat a narrow parser benchmark as an overall product comparison. It should define a shared task such as finding a locus, inspecting its transcript models, extracting a spliced sequence, associating an expression matrix, and exporting a traceable result. The observed steps and available outputs could then be reported without a subjective usability score if no participants are involved. For a claim about effort or learnability, direct task observation and a counterbalanced participant design are necessary. A fair comparison with manual file handling and browser-based workflows must use identical data and outputs and report concrete steps, intermediate files, provenance retention, and failure handling. Participant studies would be required for claims about ease of use or time saved; none was done here.

The journal's interdisciplinary remit makes the work relevant insofar as it addresses a general research infrastructure gap for plant scientists who work with locally assembled or underrepresented taxa. To support that relevance, a final paper should frame the contribution around repeatable access to dispersed public data, transparent source dependence, and verified locus-level outputs, rather than a list of software features. Our evidence justifies that technical framing but does not demonstrate increased discovery, reduced labor, or user benefit. Such outcomes require separate study designs.

Conclusion

PhytoScope provides a reusable local workflow for querying plant gene annotations and connecting them to user-supplied sequence and expression data. Exact source-derived checks across large reference annotations, selected sequence gold standards in three additional plants, and a pinned public rice-bean RNA-seq re-quantification support portability and data reconciliation for the tested releases. The results do not establish universal format support,

expression biology for the genotype-mismatched rice-bean data, user-efficiency gains, or biological discovery. The framework is best understood as a transparent, recoverable query layer over user-provided plant data, with compatibility and biological interpretation bounded by the provenance and completeness of those inputs.

Acknowledgments

[AUTHOR TO COMPLETE: acknowledge data providers, institutional support, and contributors not meeting authorship criteria.]

Author contributions

[AUTHOR TO COMPLETE using CRediT roles; verify each author's contribution and approval.]

Funding

[AUTHOR TO COMPLETE: funder names and grant numbers, or state that no specific funding was received if verified.]

Competing interests

[AUTHOR TO COMPLETE: disclose relevant interests or state no competing interests only after confirmation by all authors.]

Data availability

All software, test code, frozen sequence gold records, public expression matrix, source manifests, checksums, and evidence reports are available in the PhytoScope repository: https://github.com/omixcore/phytoscope . The evaluation inputs are identified in `docs/non-model-portability-evidence.md`, `src/test/resources/non-model/vigna-umbellata/public-expression-manifest.json`,

and the annotation validation manifests under `src/test/resources/chromosome-validation/`. Raw reads are publicly deposited under BioProject PRJNA765494; assembly and annotation files are identified by NCBI RefSeq accession GCF_029865625.1 and annotation release GCF_029865625.1-RS_2026_09. The manuscript package should be linked to the exact accepted repository commit or a versioned archival release before submission. [AUTHOR TO COMPLETE: provide permanent archive DOI and confirm repository license and data licenses.]

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Figure legends

Figure 1. PhytoScope data flow and scope. User-provided annotation, matching genome sequence, expression values, and sample metadata enter a local project. The software links files to locus search, structure view, sequence extraction, expression display, and selected analysis modules. Project state and results can be saved and reopened. Annotation-derived, sequence-derived, and user-entered information are shown as separate inputs; no function, orthology, causal regulation, or differential expression is inferred.

Figure 2. Species and data coverage in the evaluation. Four reference plant systems support annotation regression and source-to-model tuple checks. Three additional public plant assemblies support full GFF3 import/search and selected sequence checks; rice bean additionally has a matched-release transcript reference and re-quantified public reads, with a genotype mismatch. Coverage indicates tests performed, not that every data type exists for each species.

Figure 3. Exact structural tuple comparison in the reference cohort. Counts of recognized gene, transcript, exon, CDS, and UTR tuples by source inventory, including bundled Arabidopsis, full release-62 Arabidopsis, and excerpt/full annotations for rice, maize, and soybean. Total 4,641,758 tuples; all eight inventories had zero missing and zero extra tuples against independent source-derived inventories. This validates only the listed feature and relationship fields.

Figure 4. Public rice-bean expression data processing and reconciliation. Three 5-DPA RNA-seq libraries from genotype IC426787 were quantified with Salmon 2.8.0 against the GCF_029865625.1 (VRB3) RefSeq RNA FASTA and joined through its GFF3 parent IDs. All 34,914 output gene IDs matched the tested annotation. The genotype/reference mismatch limits interpretation to import and identifier compatibility. The matrix is not used for differential expression.

Figure 5. Selected independent module checks. Observed exact-match counts for promoter sequence reconstruction, domain-coordinate reconstruction, and small homology-classification panels, alongside the expression grouping arithmetic and project save/restore regression. Denominators are test-specific and should not be interpreted as comparable accuracy estimates. Protein similarity panels represent RBCS-family smoke tests, not orthology validation.